THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — SCIENCE · MEDIUM LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A student tests the effect of the mass of a bob on the time period of a pendulum. Which is the dependent variable? 1

(a) place of testing **(b)** mass of the bob
(c) length of the string **(d)** time period

2. A microscope has a 10x eyepiece and a 40x objective. A cell appears 2 mm wide in the image. The actual width is: 1

(a) 0.005 mm **(b)** 800 mm **(c)** 0.5 mm **(d)** 0.05 mm

3. A person has a fever, a rash and a headache. Which of these is a sign, not a symptom? 1

(a) headache **(b)** body ache
(c) rash seen by the doctor **(d)** feeling weak

4. A student connects a bulb, a switch and a cell but the bulb does not glow even when the switch is closed. The bulb holder is loose. The fault is a: 1

(a) overload **(b)** magnetic effect
(c) short circuit **(d)** break in the circuit

5. A block is pulled with 4 N but does not move. The friction acting on it is: 1

(a) 8 N **(b)** 4 N **(c)** 2 N **(d)** 0 N

6. A force of 240 N acts on an area of 0.4 square metre. The pressure is: 1

(a) 600 Pa **(b)** 60 Pa **(c)** 240 Pa **(d)** 96 Pa

7. A crystal of potassium permanganate is dropped into water in two beakers, one hot and one cold. The colour spreads faster in the hot beaker because: 1

(a) cold water is thicker
(b) crystal dissolves into gas
(c) particles move faster at higher temperature
(d) water gets heavier

8. Which of the following pairs contains only elements? 1

(a) $NaCl$ and Fe **(b)** O_2 and Fe
(c) H_2O and O_2 **(d)** CO_2 and N_2

9. 40 g of salt is added to 160 g of water. The mass percentage of the solution is: 1

(a) 25 per cent **(b)** 10 per cent **(c)** 40 per cent **(d)** 20 per cent

10. A ray strikes a plane mirror making an angle of 25 degrees with the mirror. The angle of reflection is: 1

(a) 50 degrees **(b)** 155 degrees **(c)** 65 degrees **(d)** 25 degrees

11. A vertical stick of 1 m casts a shadow of 1 m long at some time. The Sun's elevation at that time is about: 1

(a) 60 degrees **(b)** 30 degrees **(c)** 90 degrees **(d)** 45 degrees

12. In the food chain: phytoplankton, zooplankton, small fish, big fish, the secondary consumer is: 1

(a) phytoplankton **(b)** small fish **(c)** big fish **(d)** zooplankton

13. If the radius of the Earth is about 6400 km, the thickness of the crust (about 35 km) is: 1

(a) equal to the radius **(b)** less than 1 per cent of the radius
(c) about 50 per cent of the radius **(d)** about 25 per cent of the radius

14. After heating, the temperature readings are 40, 60, 80 and 100 degrees C at times 1, 2, 3 and 4 min. The rate of rise is: 1

(a) 10 degrees C per minute **(b)** 40 degrees C per minute
(c) 100 degrees C per minute **(d)** 20 degrees C per minute

15. A tissue differs from an organ because a tissue: 1

(a) is an organism **(b)** forms systems
(c) has many types of tissues **(d)** has similar cells doing one job

16. The correct sequence of events in infection by a pathogen is: 1

(a) symptoms, entry, multiplication **(b)** symptoms, multiplication, entry
(c) entry, multiplication, symptoms **(d)** multiplication, symptoms, entry

17. A toaster, an iron and a heater use the same effect of current, which is: 1

(a) light **(b)** heating **(c)** magnetic **(d)** chemical

18. A glass rod is rubbed with silk and brought near a suspended charged plastic rod, which is repelled. The plastic rod has: 1

(a) negative charge **(b)** no charge **(c)** magnetism **(d)** positive charge

19. Air pressure inside a bicycle tyre is greater than outside because: 1

(a) rubber is thick **(b)** the valve is open
(c) tyre is hot **(d)** more air is pumped in

20. A steam burn is worse than a burn from boiling water at the same temperature mainly because steam: 1

(a) gives out extra heat when it condenses on the skin
(b) is heavier than water
(c) has a larger mass
(d) is at a higher temperature than boiling water

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Why are the independent and dependent variables different in an investigation? 2

22. Why are bacteria placed in a different group from plants and animals? 2

23. Why is stagnant water a health risk? Name two diseases that result. 2

24. A bulb does not glow in a circuit. List two possible faults and how you would check them. 2

25. A trolley is pushed with 15 N and the friction is 6 N. Find the net force and say in which direction it acts. 2

26. A box exerts a force of 80 N on a floor over 0.2 square metre. Find the pressure. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. A student plans to test whether sugar dissolves faster in warm water. List the variable to change, the variable to measure and three things to keep same. 3

28. A cell shows a thick wall, large vacuole and green bodies. Identify the cell, name the green bodies and state their function. 3

29. Explain how a vaccine produces immunity. Why are booster doses needed? 3

30. A student made an electromagnet but it can't pick up pins. List three possible reasons and remedies. 3

31. A box of mass 10 kg is on a floor. A spring balance shows 25 N just before it starts moving and 18 N while sliding. Explain the readings. 3

OR

A girl of weight 400 N stands on one foot of area 0.02 square metre. Find the pressure and compare with standing on two feet. 3

32. Explain why solids retain their shape, liquids flow and gases spread using particle ideas. 3

OR

A student mixed iron filings and sulphur and heated them. Write three observations and the inference. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Plan an investigation to find out which of three liquids (water, milk, oil) cools fastest. Mention what you measure, what you keep constant, and the procedure. 5

34. Draw up a comparison between the plant cell, animal cell and bacterial cell in a table in terms of nucleus, cell wall, chloroplast, vacuole and size. 5

35. Explain the chain of infection of a communicable disease using cholera as an example, and show where it can be broken. 5

36. Describe an activity to show that a current-carrying wire behaves like a magnet. Mention the observations when the current is reversed and when the compass is placed in different positions. 5

OR

A student pulls a block with a spring balance on three surfaces: polished table (2 N), wooden floor (5 N) and carpet (9 N). Explain the observations and draw a conclusion. How will the readings change if a 1 kg mass is added? 5

37. A tractor with weight 36000 N has four wheels each in contact with 0.3 square metre of ground. Find the pressure on the ground and compare with a person of weight 600 N standing on feet of area 0.05 square metre in total. 5

OR

Describe an experiment to show that particles of matter are continuously moving and have spaces between them, using KMnO4 and water. 5

38. Describe the properties of the iron-sulphur mixture and iron sulphide, and conclude what is seen about mixtures and compounds. 5

OR

Describe an experiment to find out how the solubility of salt and of sugar changes with temperature. State the variable, procedure, table and conclusion. 5

END OF QUESTION PAPER **Total: 80 Marks**

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